Earnings Surprise, Firm Valuation, and Market Efficiency: Evidence from China

Corporate Finance Team Project

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FIN803 Corporate Finance

Motivation

- Earnings news should update firm value through expected future cash flows.
- In semi-strong efficiency, public guidance information should be priced quickly.
- China A-share offers a rich setting for testing post-announcement price adjustment.
- Corporate Finance relevance: valuation, information processing, and cost of capital.
- This project directly maps to FIN803 topics: stock valuation, pricing of risk, CAPM-based abnormal return measurement, and cost-of-capital intuition.

Research Question

Core Question

Do earnings-related guidance events predict positive post-announcement abnormal returns in China A-share stocks?

- Main focus: event occurrence effect vs surprise magnitude ranking.
- Secondary focus: whether cross-sectional ranking by the current ES proxy is stable.
- Final framing prioritizes economic interpretation over trading optimization.

Firm Valuation and Earnings

- Firm value reflects discounted expected future cash flows.
- Earnings guidance updates beliefs about future profitability.
- Valuation impact depends on both **earnings level** and **expectation gap**.
- Surprise component matters because prices react to *new information*, not raw earnings alone.

Market Efficiency

- Semi-strong EMH: prices should rapidly incorporate public information.
- Under full efficiency, post-announcement abnormal return should be near zero.
- Persistent positive CAR after guidance implies delayed adjustment.
- This is a direct empirical test of financial theory, not a strategy claim.

Abnormal Returns and CAPM

- Expected return benchmark is linked to systematic risk (beta).
- CAPM gives cost-of-equity intuition for normal required return.
- Abnormal return isolates pricing response beyond market-wide movement.

$$AR_{i,t} = R_{i,t} - R_{m,t}$$

- If earnings news has information value, event-window CAR can deviate from zero.

PEAD Intuition

- PEAD: prices may continue drifting after earnings-related announcements.
- Possible drivers: investor attention limits, processing frictions, uncertainty.
- We test delayed reaction, but do **not** assume strong monotonic PEAD ex ante.
- Our evidence is interpreted as event-level underreaction with proxy limitations.

Data Description

- Market: China A-share listed firms.
- Sample period: 2020-01-01 to 2026-04-16.
- Expanded sample: 800 stocks, 4,980 usable events.
- Data sources: Tushare (primary) with backup/cross-check pipeline.

Event Definition

- **guidance_initial**: first valid guidance for a report period.
- **guidance_upward_revision**: later guidance with both bounds revised upward.
- Event trading date: first trading day after announcement.
- One event timeline per report-period logic to avoid duplicate counting.

Filtering and Tradability

- Exclude ST stocks.
- Exclude firms with fewer than 120 trading days since listing.
- Exclude suspended/non-tradable event-day observations.
- Apply liquidity screen (20-day turnover threshold) to keep practical tradability.
- Purpose: reduce microstructure noise and improve inference quality.

Return Measurement

- Compute abnormal return daily: $AR_{i,t} = R_{i,t} - R_{m,t}$.
- Event-window outcomes:
  - $CAR20 = \sum_{t=1}^{20} AR_{i,t}$
  - $CAR60 = \sum_{t=1}^{60} AR_{i,t}$
- CAR60 is the main metric for delayed adjustment (underreaction horizon).

Earnings Surprise Proxy

- Proxy-based surprise:

$$ES_{\text{main}} = \text{guidance_yoy_midpoint} - \text{analyst_consensus_yoy_proxy}$$

- Consensus term is an approximation, not true analyst consensus.
- This measurement limitation is central to interpretation.

Interpretation Caution

Unstable ES ranking power can arise from expectation-measurement noise.

ES Grouping

- **positive ES**: $ES_{\text{main}} > 0$
- **moderate ES**: 50th–80th percentile within event type
- **extreme ES**: top 20% within event type
- Goal: compare economically meaningful groups instead of forcing strict monotonic ranking.

Empirical Strategy

- Event-study framework around guidance announcements.
- Compare average CAR across event categories and ES signal groups.
- Emphasis on **event occurrence effect** over fine ES ranking.
- Robustness: expanded sample and tradability filters.

Regression Model

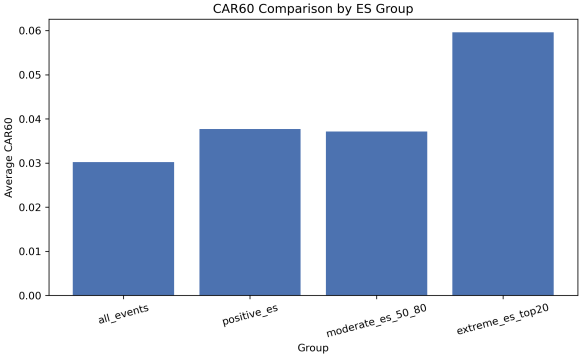
- Baseline linear specification with robust standard errors:

$$CAR60_i = \alpha + \beta_1 \cdot signal_i + \beta_2 \cdot \log(size_i) + \beta_3 \cdot beta_i + \varepsilon_i$$

- Two final signals:
  - $signal =$ moderate_positive_ES_dummy
  - $signal =$ event_type_dummy (upward revision $= 1$)
- Purpose: test incremental explanatory power of event-related information.

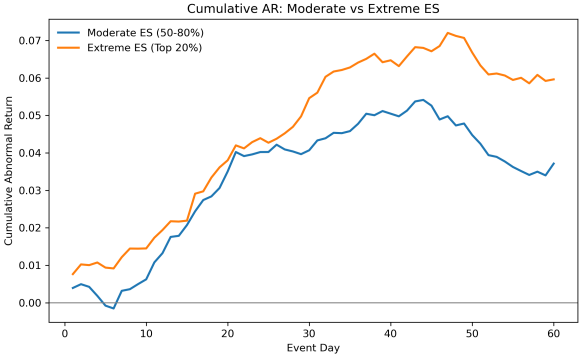
Main Event-Level Results

- All-events average $CAR60 \approx 3.02\%$.
- Positive-ES events: $CAR60 \approx 3.77\%$.
- Guidance events are linked to positive post-announcement abnormal returns.



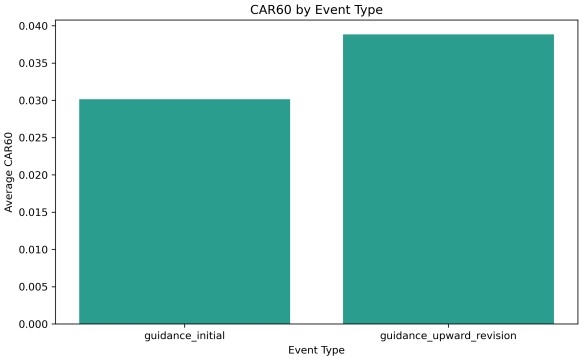
ES Group Comparison

- Moderate ES (50–80%): $CAR60 = 3.715\%$
- Extreme ES (Top 20%): $CAR60 = 5.964\%$
- ES ranking is not stable across specifications.
- Magnitude ordering alone is not robust.



Regression Results

- Moderate ES dummy: 0.0092, $p = 0.254$ (positive, insignificant).
- Event-type dummy: 0.0009, $p = 0.980$ (insignificant).
- The chart shows a higher raw CAR60 mean for revisions, but the conditional difference is not robust after controls.
- Beta remains significant for CAR60.
- Event signal exists, linear cross-section is weak.



Corporate Finance Interpretation

- Earnings-related disclosures affect valuation through expectation updates.
- Positive post-announcement CAR suggests delayed information incorporation.
- Underreaction is present, but not in a clean linear ES-magnitude pattern.
- CAPM controls do not fully absorb short-horizon earnings-announcement effects.
- Measurement of expected earnings is critical for theory-consistent testing.

Limitations and Conclusion

- Main limitation: proxy-based expected earnings (no full analyst consensus feed).
- Result strength: robust event-level evidence $>$ unstable ranking evidence.
- Practical takeaway: event occurrence is more reliable than surprise magnitude ordering.
- Final conclusion: earnings information matters for firm value, but market adjustment is imperfect.
- Next step: improve expectation measurement to sharpen cross-sectional tests.